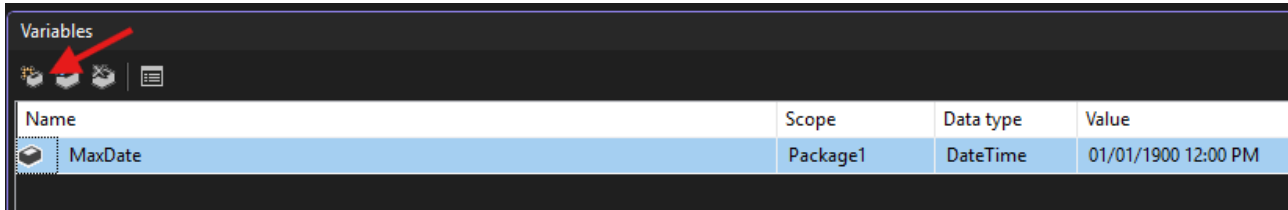


Lab3:

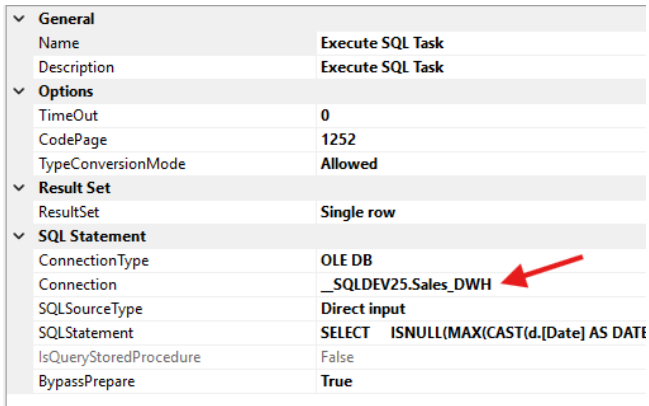
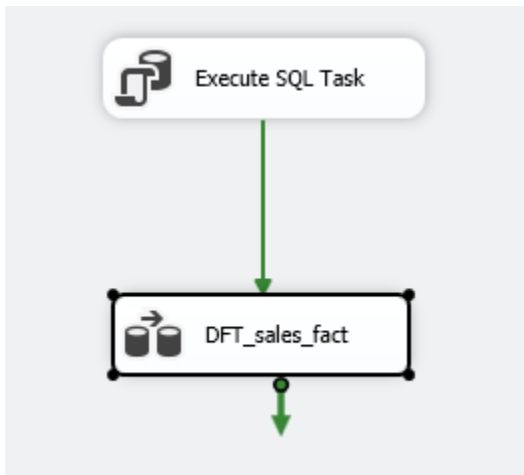
Prevent duplication in the fact table using incremental load with variable holding the max datetime

Step1: Create a variable give name, type: DateTime and initial value for first integration



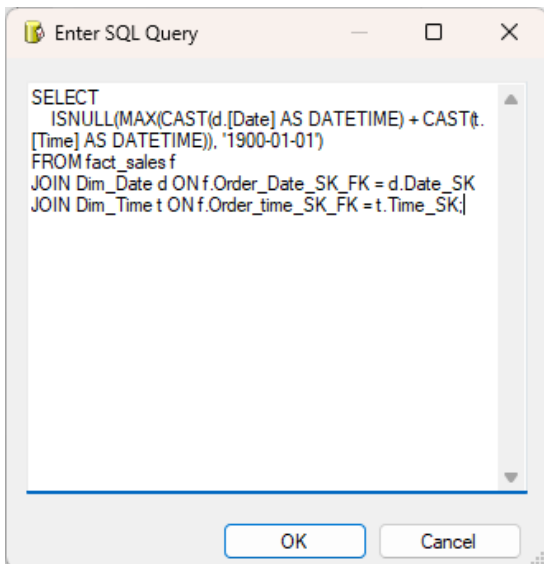
Name	Scope	Data type	Value
MaxDate	Package1	DateTime	01/01/1900 12:00 PM

Step2: add Execute SQL Task before the Data Flow Task and make sure the connection is to the DWH and the ResultSet to Single row



Execute SQL Task	
Name	Execute SQL Task
Description	Execute SQL Task
Options	
TimeOut	0
CodePage	1252
TypeConversionMode	Allowed
Result Set	
ResultSet	Single row
SQL Statement	
ConnectionType	OLE DB
Connection	_SQLDEV25.Sales_DWH
SQLSourceType	Direct input
SQLStatement	SELECT ISNULL(MAX(CAST(d.[Date] AS DATE
IsQueryStoredProcedure	False
BypassPrepare	True

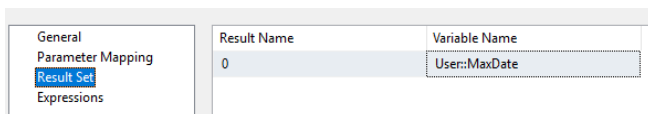
And then in the SQL Statement select the max Date and Time from the fact table



```
SELECT
  ISNULL(MAX(CAST(d.[Date] AS DATETIME) + CAST(t.
[Time] AS DATETIME)), '1900-01-01')
FROM fact_sales f
JOIN Dim_Date d ON f.Order_Date_SK_FK = d.Date_SK
JOIN Dim_Time t ON f.Order_time_SK_FK = t.Time_SK;
```

OK Cancel

Then in the Result Set add the new result with name 0 and select the created variable



Result Name	Variable Name
0	User::MaxDate

Step3: Inside the Data Flow Task change the SQL Command to have this where condition

SQL command text:

```
select
    o.OrderID,
    od.OrderDetailsID,
    o.CustomerID,
    o.SalesmanID,
    od.ProductID,
    od.Quantity,
    od.TotalPrice,
    o.OrderDate
from [Order] o
join OrderDetails od
on o.OrderID = od.OrderID
where o.OrderDate > ?
```

Parameters...

Build Query...

Browse...

Parse Query

Then choose parameters to choose a variable to replace the (?) that is (User::MaxDate)

Mappings:		
Parameters	Variables	Param direction
Parameter0	User::MaxDate	Input

And Congrats it's all Done just make sure that Destination mapping didn't change